ARIS

AI-Assisted Smart Helmet Communication System

CMPE 491- Senior Project

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Motivation & Problem Definition

- Communication is difficult in dynamic field and military environments.
- Handheld communication devices limit mobility and situational awareness.
- Manual operation is unsafe under stress, noise and physical constraints.
- Fast, reliable and hands-free command transmission is required.

Project Overview

01

ARIS is an AI-assisted smart helmet communication system

02

Uses multimodal inputs: gesture, voice and facial expressions

03

Converts user inputs into predefined commands

04

Enables secure communication with a central system

Project Objectives

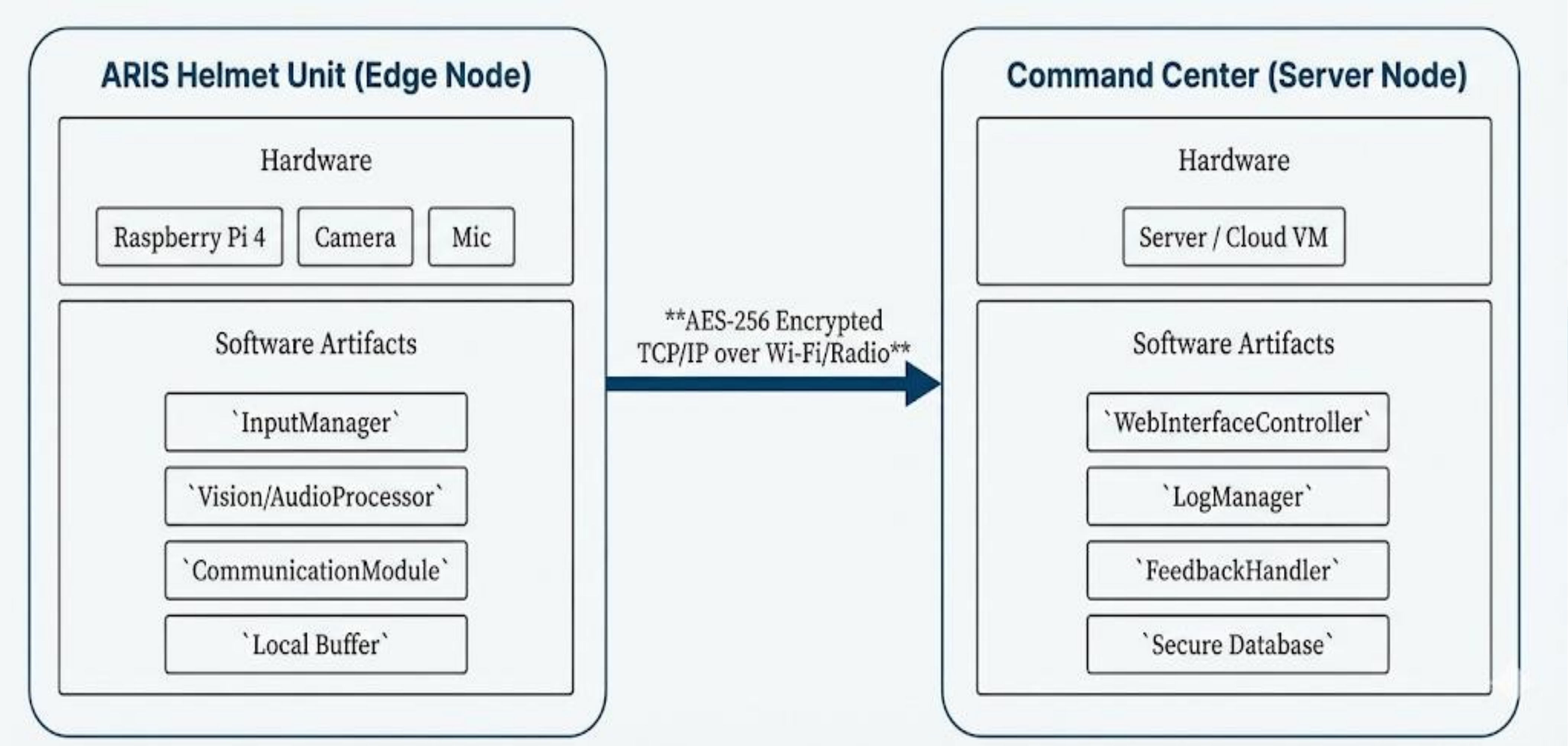
- Provide hands-free communication in field environments
- Improve user mobility and operational safety
- Reduce communication delay using AI-based recognition
- Design a modular and scalable system
- Demonstrate feasibility through a functional prototype

Project Scope & Assumptions

Project Scope
Gesture, voice and facial expression recognition
AI-based command interpretation
Secure wireless command transmission
Web-based monitoring interface
Prototype-level implementation

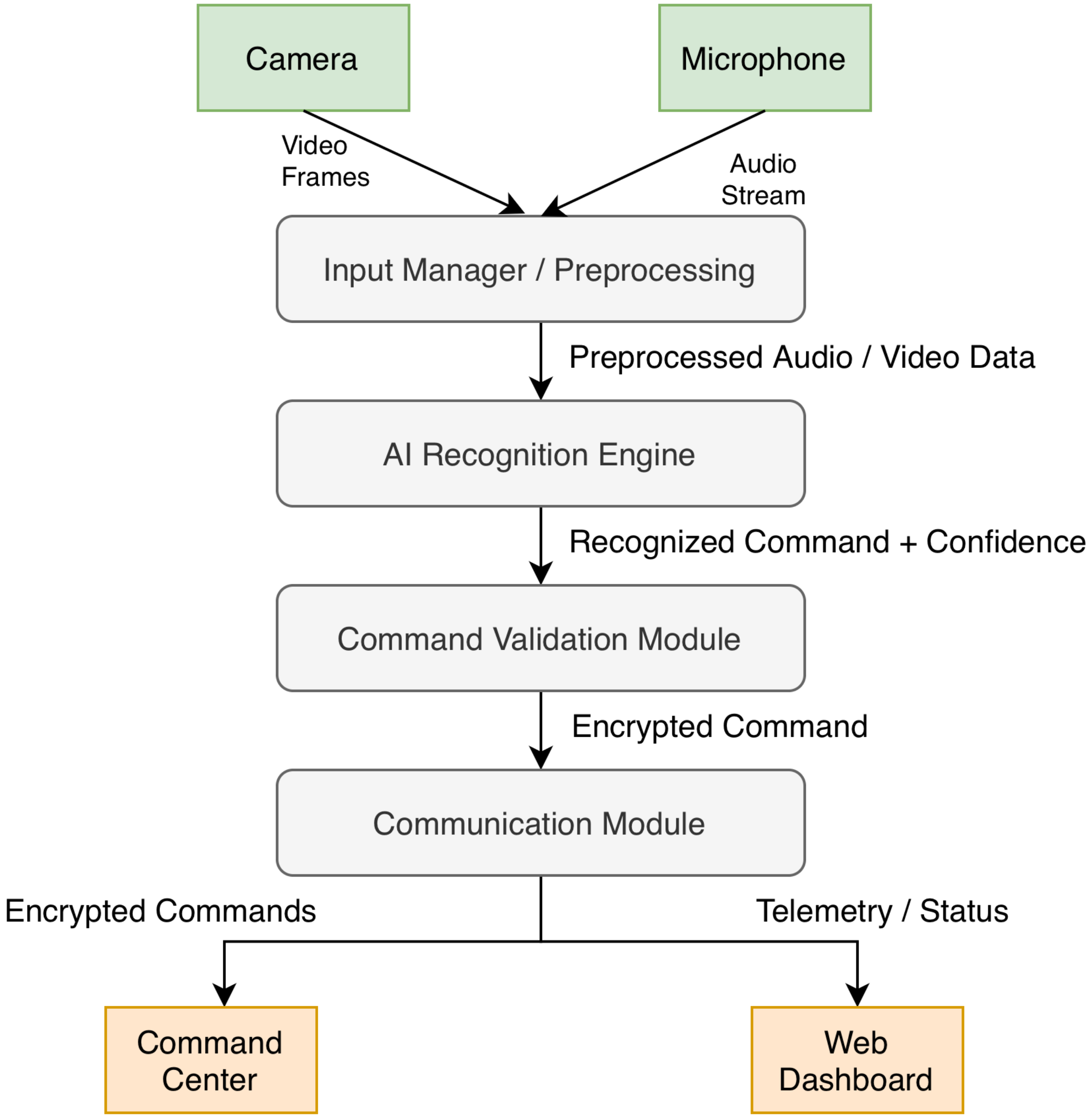
Assumptions
Operates in a simulated environment
Users are trained on predefined commands
Tested under moderate conditions

System Design



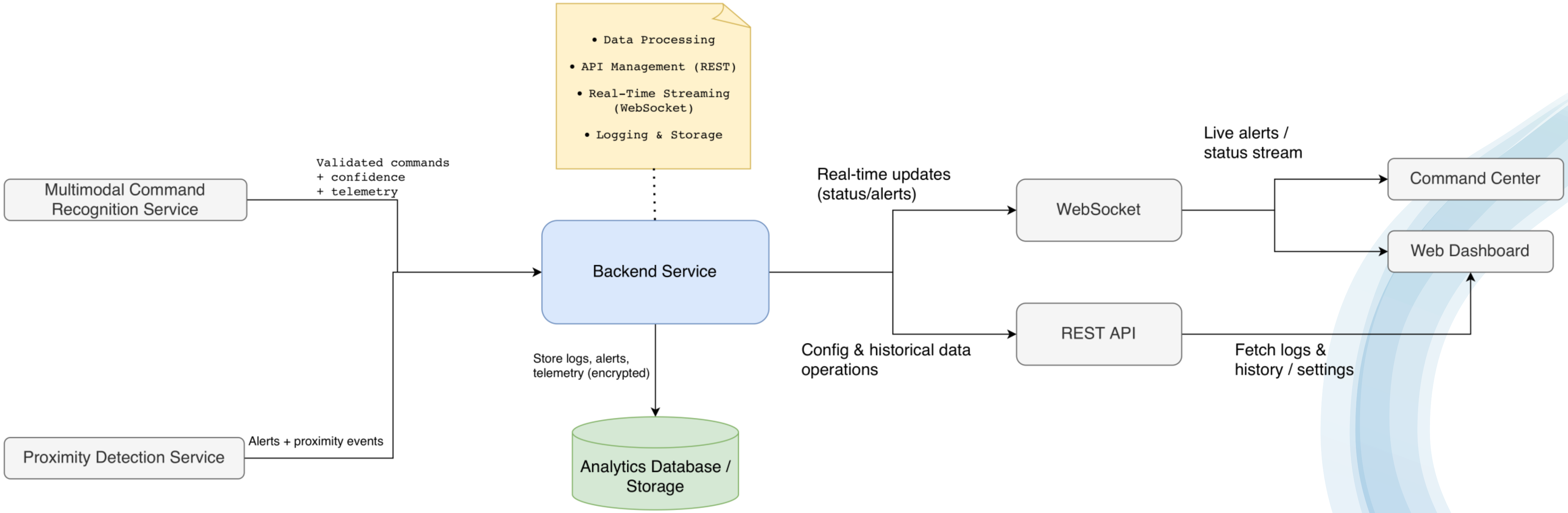
Subsystem Design

- The edge-level processing on the ARIS Helmet, where audio–video data is preprocessed, analyzed by AI modules, validated, and securely transmitted.



Subsystem Design

- The server-side architecture, responsible for real-time streaming, command handling, logging, analytics, and dashboard interaction.



ARIS Pipeline Flow

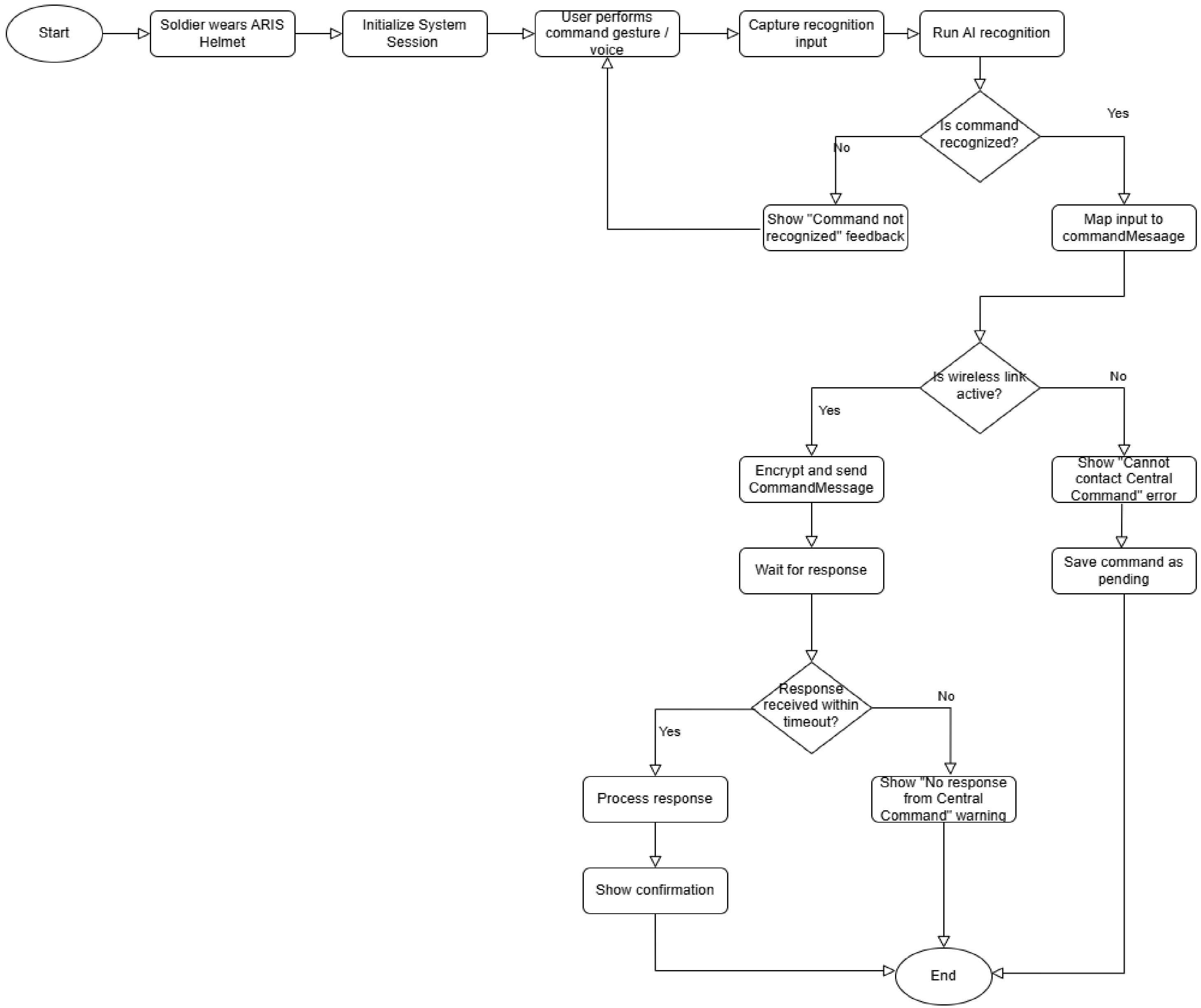
Input Capture: The camera and microphone capture user gestures or voice commands.

Edge Recognition: The Raspberry Pi 4 processes these inputs using AI to identify valid commands.

Secure Transmission: Validated data is encrypted with AES-256 and sent via Wi-Fi/Radio.

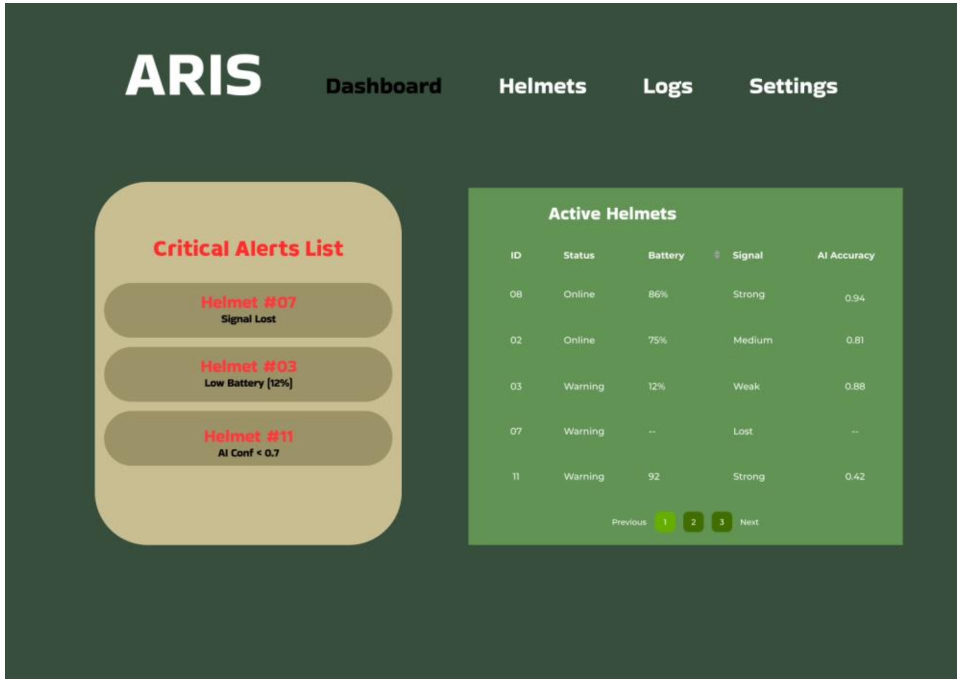
Central Management: The Command Center logs the data and stores it in a secure database.

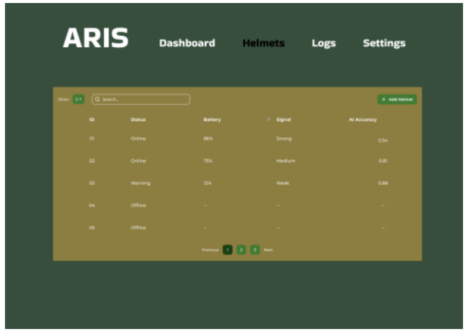
Feedback Loop: A response is sent back to the helmet to display a final confirmation to the user.

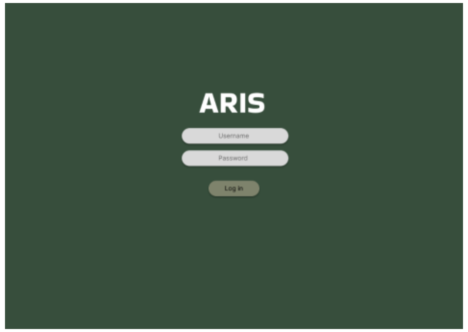


Mockups

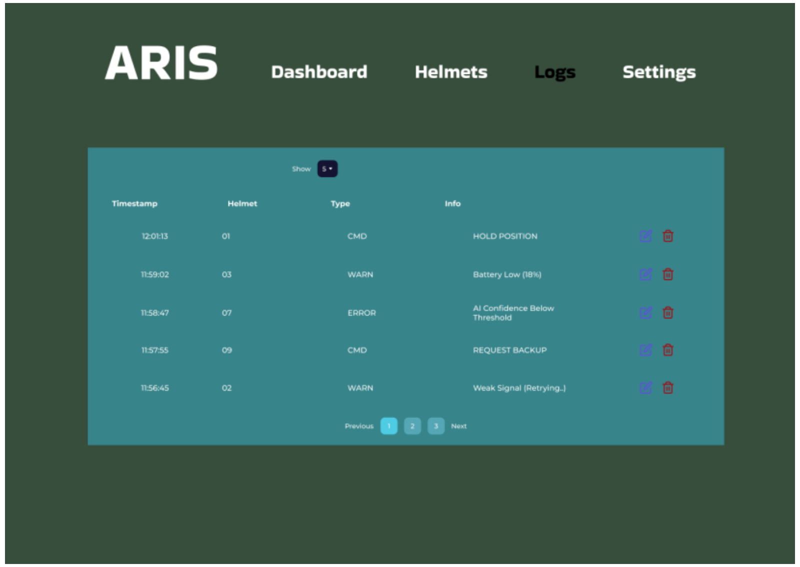
- Clean and Focused
- Real-Time Monitoring
- Instant Decision Support
- Tactical Interface

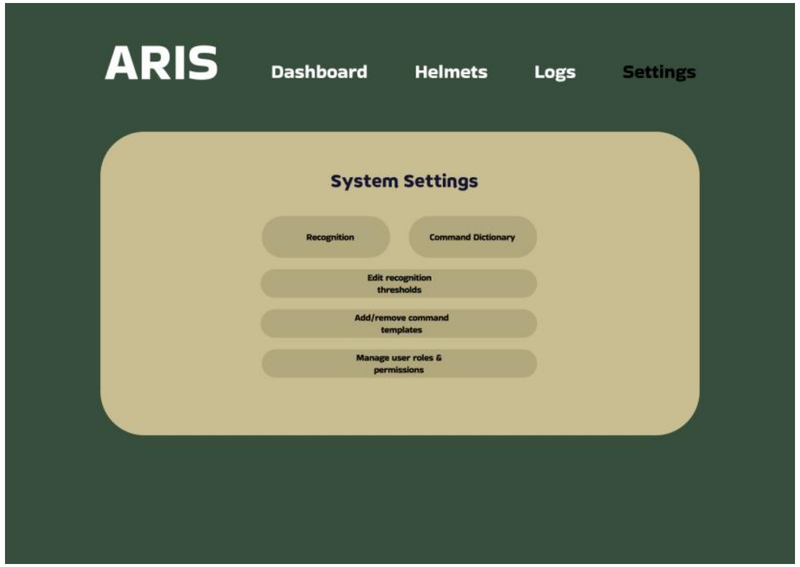


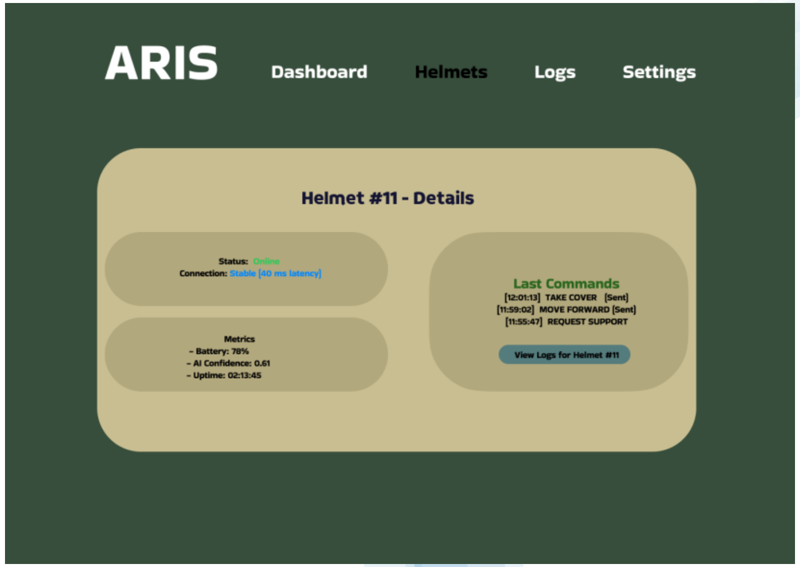




Mockups


ARIS
Dashboard
Helmets
Logs
Settings
Show 5
Timestamp
Helmet
Type
Info
12:01:13 01 CMD HOLD POSITION
11:59:02 03 WARN Battery Low (18%)
11:58:47 07 ERROR AI Confidence Below Threshold
11:57:55 09 CMD REQUEST BACKUP
11:56:45 02 WARN Weak Signal (Retrying..)
Previous 1 2 3 Next


ARIS
Dashboard
Helmets
Logs
Settings
System Settings
Recognition
Command Dictionary
Edit recognition thresholds
Add/remove command templates
Manage user roles & permissions


ARIS
Dashboard
Helmets
Logs
Settings
Helmet #11 - Details
Status: Online
Connection: Stable [40 ms latency]
Metrics
- Battery: 78%
- AI Confidence: 0.61
- Uptime: 02:13:45
Last Commands
[12:01:13] TAKE COVER [Sent]
[11:59:02] MOVE FORWARD [Sent]
[11:55:47] REQUEST SUPPORT
View Logs for Helmet #11

Sample Test Video

Future Implementations

Algorithm Optimization

Enhanced Network

Web Interface Performance Optimization

Verification and Validation

Thank You!